

The gapped coupling set: finite-volume continuity and conditional spectral limits

Later-note pointers. Openings, §1 corrects the all-coupling requirement; Lieb–Robinson, §§1–2 and 5 supplies the interaction-picture locality argument while leaving gap stability open; three continuum limits, §5 states the surviving-spectral-weight conditions missing from the original argument.

Correction (2026-10-02). The original title and lead identified T2' with absence of a zero-temperature bulk transition and, on a putative second-order branch, uniqueness of the continuum limit. Those equivalences and the original Proposition 2/Corollary 3 are **withdrawn**: a subsequence of closing finite-volume gaps does not establish an infinite-volume theory, its vacuum multiplicity or its local spectral threshold. The two proposed mechanisms remain possible interpretations, not an exhaustive proved dichotomy. The compact $U(1)$ claim of a second-order Wilson transition with continuously vanishing confined-phase mass and a Maxwell continuum limit at its endpoint is also withdrawn. The corrections below retain finite-volume continuity, replace Proposition 2 with a conditional spectral statement, and separate lattice-action evidence from the Kogut–Susskind operator defined here.

Define, for the Kogut–Susskind Hamiltonian of the obligations map, with gauge group G , lattice spacing a and N_s sites per side,

$$c(g) = \liminf_{N_s \rightarrow \infty} \delta(g; N_s, G), \quad \mathcal{G} = \{g > 0 : c(g) > 0\}, \quad \Delta_{a,L} = \frac{\hbar c}{a} \delta(g; N_s, G), \quad L = aN_s.$$

Here $c(g)$ is a dimensionless lower-gap function; the constant c in $\hbar c/a$ is the speed of light. The volume limit is taken at fixed a and g . To retain the obligations map's convention, δ is the gap on the **full link Hilbert space**. Its restriction to the Gauss-law space has a gap $\delta^{\text{phys}} \geq \delta$; closing of δ need not even close δ^{phys} . An analogous gapped set can be defined using δ^{phys} , but the two sets have not been identified here.

T1 gives positivity at every finite volume; T2 gives $[g_0(G), \infty) \subset \mathcal{G}$; and T2', as an all-coupling finite-volume lower-bound statement, is exactly $\mathcal{G} = (0, \infty)$. Proposition 1 proves finite-volume continuity. Proposition 2 identifies the slowest temporal correlation decay with the spectral threshold **assuming a specified limiting vacuum theory**; it does not identify that threshold with $c(g)$. Global openness and closedness of $\mathcal{G}$ remain open. If openness is established, closedness and the known nonemptiness would imply T2' by connectedness. This is a sufficient route to a lattice statement, not an equivalence with the continuum mass-gap conjecture or with phase-transition absence.

1. The gap at fixed lattice is continuous in the coupling

Proposition 1. For fixed a , $N_s \geq 2$ and a nontrivial compact connected Lie group G , the map $g \mapsto \delta(g; N_s, G)$ is continuous and strictly positive on $(0, \infty)$, and locally piecewise real-analytic. The same holds for the physical gap.

Proof. Write

$$h(g) := \frac{aH(g)}{\hbar c} = \frac{g^2}{2}K + \frac{2}{g^2}V,$$

with $K = \sum_{\ell} (-\Delta_{\ell}) \geq 0$ having compact resolvent and $V = \sum_p (N - \text{Re tr } U_p)$ bounded, with $0 \leq V \leq 2N|\mathcal{P}|$ for $SU(N)$ in the fundamental representation. For a general compact connected group, use the fixed faithful representation and replace N by its dimension. In a complex neighbourhood of any $g > 0$, this is a holomorphic family of type (A) on the common domain $D(K)$. For real g it is self-adjoint with compact resolvent. The local analytic eigenvalue-branch theorem for a self-adjoint analytic family (Kato/Rellich analytic perturbation theory, standard theorem used here) supplies finitely many real-analytic branches in an isolated spectral

cluster. The ground branch is simple by T1. In a neighbourhood of each g , the first excited eigenvalue is the minimum of the finitely many branches in its cluster, hence continuous. Distinct analytic branches have isolated crossings locally; identically coincident branches do not affect this conclusion. Thus the gap is locally piecewise analytic and continuous. T1 supplies its strict positivity. The gauge projection reduces both K and V , so the same argument applies on the fixed physical subspace. $\square$

Correction (2026-10-02). The original min-max explanation did not prove continuity: an infimum over subspaces of continuous functions need not be continuous. The analytic-branch argument above replaces it; no simplicity of the first excited eigenvalue is assumed.

Finite-volume positivity supplies no positive volume-uniform lower bound. In particular $c(g)$ need not be continuous, and no regularity of $\mathcal{G}$ follows just from Proposition 1.

2. What closing finite-volume gaps does and does not imply

If $c(g) = 0$, then $\delta(g; N_s, G) \rightarrow 0$ along a subsequence of volumes. This is a statement about eigenvalues of different operators, not about the spectrum of a constructed infinite-volume operator. To discuss correlation length, first assume a limiting vacuum representation of the local gauge-invariant observable algebra, with cyclic vacuum Ω , nonnegative dimensionless Hamiltonian h_∞ , and $\ker h_\infty = \mathbb{C}\Omega$. The physical time generator is $H_\infty = (\hbar c/a)h_\infty$. Existence of this representation and convergence of the relevant correlators are extra hypotheses here.

For each bounded local gauge-invariant observable A , set $A_0 = A - \langle \Omega, A\Omega \rangle I$ and define its finite positive spectral measure and temporal connected autocorrelation by

$$\mu_A(B) = \langle A_0\Omega, \mathbf{1}_B(h_\infty)A_0\Omega \rangle, \quad C_A(s) = \int_{[0,\infty)} e^{-s\lambda} d\mu_A(\lambda), \quad s = \frac{ct_E}{a}.$$

For $\mu_A \neq 0$ define $r_A = \inf \text{supp } \mu_A$ and

$$r = \inf_{A: \mu_A \neq 0} r_A, \quad \xi_t = \frac{a}{r},$$

with $\xi_t = \infty$ when $r = 0$. This takes the **slowest rate over all local gauge-invariant observables**, rather than the rate of an arbitrarily chosen observable. It is a temporal correlation length; no equality with a spatial correlation length at fixed Hamiltonian lattice spacing is asserted.

Proposition 2 (conditional replacement). In the vacuum representation just specified, with a nonzero excited subspace,

$$m_{\text{lat}} := \inf \sigma(h_\infty|_{\Omega^\perp}) = r, \quad \Delta_\infty = \frac{\hbar c}{a} m_{\text{lat}} = \frac{\hbar c}{\xi_t}.$$

For every nonzero μ_A , $-\lim_{s \rightarrow \infty} s^{-1} \log C_A(s) = r_A$. These assertions concern the assumed limit theory; they make no claim that $m_{\text{lat}} = c(g)$.

Proof. Positivity of the spectral measure gives $C_A(s) \leq \mu_A([0, \infty))e^{-sr_A}$. For every $\varepsilon > 0$, the definition of support gives positive weight $w_\varepsilon = \mu_A([r_A, r_A + \varepsilon]) > 0$, whence $C_A(s) \geq w_\varepsilon e^{-s(r_A + \varepsilon)}$. Taking logarithms, then $s \rightarrow \infty$ and $\varepsilon \downarrow 0$, proves the rate formula. Centered local vectors are dense in $\Omega^\perp$ by cyclicity of Ω for the local algebra. All their spectral supports lie above m_{lat} , so $r \geq m_{\text{lat}}$. For every $\varepsilon > 0$ the spectral projection on $[m_{\text{lat}}, m_{\text{lat}} + \varepsilon]$ in $\Omega^\perp$ is nonzero. Density implies that it has nonzero action on some centered local vector. That vector has $r_A < m_{\text{lat}} + \varepsilon$, proving the reverse inequality after $\varepsilon \downarrow 0$. The physical-unit identity follows from the definition of H_∞ . $\square$

An infimum of rates equal to zero need not be attained by a single observable: $\xi_t = \infty$ need not mean that *one* observable has slower-than-exponential decay. It can mean that there is no common positive exponential rate for the entire local algebra.

Why the original implication fails. Even convergence of an observable's spectral measures does not preserve their lowest support points. With d_x denoting unit point mass at x , consider

$$\mu_n = (1 - e^{-n})d_1 + e^{-n}d_{1/n}.$$

Its threshold tends to zero, but $\mu_n \Rightarrow d_1$, whose threshold is one. The soft mode loses all its observable weight. This is a spectral-measure illustration, not a counterexample constructed for the fixed Kogut–Susskind family. It identifies information that finite-volume eigenvalues alone lack. Nonlocal sectors and the full-space/physical-space distinction add further obstacles.

A valid sufficient implication is available **with surviving weight**. Suppose centered finite-volume local spectral measures $\mu_n \Rightarrow \mu_A$ in the representation above and, for every $\eta > 0$, there is $w_\eta > 0$ with $\mu_n([0, \eta/2]) \geq w_\eta$ eventually. The closed-set inequality gives $\mu_A([0, \eta/2]) \geq w_\eta$. Uniqueness of the vacuum and centering give $\mu_A(\{0\}) = 0$, so $\mu_A((0, \eta)) > 0$ and $m_{\text{lat}} = 0$. These are precisely the extra convergence and weight hypotheses of three continuum limits, §5 (reading: passage). No such estimates are proved here for arbitrary g .

Correction (2026-10-02): phase interpretation. Vacuum coexistence with small finite-volume tunnelling splittings and a gapless local phase are possible scenarios, not an exhaustive consequence of $c(g) = 0$. Distinct infinite-volume ground states may persist throughout a symmetry-broken interval, with a positive excitation gap in each selected vacuum representation. A massless phase can likewise occupy an interval. Neither every degeneracy nor every gapless coupling is an isolated first- or second-order transition; neither an exponential volume law for tunnelling nor transition order was proved by the original argument.

Corollary 3 (original equivalence withdrawn, 2026-10-02). There is no proved equivalence here between $\mathcal{G} = (0, \infty)$ and absence of bulk phase transitions. T2' means only the all-coupling lower-bound property defined in the lead. Identifying it with phase behaviour needs a limit theory, a specified vacuum/sector and spectral comparison.

Precision, 2026-09-23, corrected 2026-10-02. The openings note, §1 (reading: passage) records Wilson-action numerical evidence for first-order bulk transitions for $SU(N \geq 5)$, with source Rindlisbacher–Rummukainen–Salami. It does not prove a hole in the $\mathcal{G}$ of this KS Hamiltonian: the Euclidean Wilson operator differs, and the spectral comparison is missing. Such intermediate-coupling lattice phenomena do not by themselves obstruct a continuum limit along $g \rightarrow 0$. All-coupling T2' is stronger than the weak-trajectory gap and scaling control needed for T3, together with continuum construction and nontriviality T4.

3. What a second-order point could mean

Correction (2026-10-02). A divergent correlation length is a necessary scaling signal for a continuum limit with a fixed nonzero physical mass, not a proof that a continuum theory exists. The original assertion that a second-order point automatically produces a continuum limit, and the equivalence with uniqueness of the asymptotically free limit, are withdrawn.

If a positive dimensionless local spectral mass $m_{\text{lat}}(g)$ tends to zero as $g \rightarrow g_* > 0$, one can choose $a(g) = m_{\text{lat}}(g)/M$ to keep the inverse-length scale $M > 0$ fixed. Then $\Delta(g) = \hbar c M$ and $\xi_t/a = 1/m_{\text{lat}} \rightarrow \infty$. This choice supplies scale matching only. Convergence of renormalized correlators, positivity, axioms and nontriviality remain to be proved. Moreover, diverging length alone does not specify transition order.

An asymptotically free trajectory instead approaches $g = 0$ with $a\Lambda_{\text{lat}}(g) \rightarrow 0$ as described in the obligations map, §5. A possible finite- g_* scaling limit would be a different trajectory; T2' alone classifies neither continuum limits nor universality classes. Numerically, the fundamental–adjoint plane of $SU(3)$ exhibits scalar softening consistent with a critical endpoint and a ϕ^4 continuum interpretation (Heller 1995, reading: abstract). This is evidence for that interpretation, not a constructed scalar continuum theory or a theorem for the Wilson axis.

4. The abelian case: separate action, phase and transition order

The strong-coupling theorem applies to compact $U(1)$: in the normalization $-\Delta$ has eigenvalues n^2 , so $C_2(R_{\min}) = 1$ and the defining representation has dimension one. The T2 note (reading: full-read) gives the KS bound

$$\delta(g; N_s, U(1)) \geq \gamma \frac{g^2}{2}, \quad g^4 \geq \frac{24}{\beta_*},$$

with existential constants $\gamma, \beta_* > 0$ independent of volume. This proves a strong-coupling half-line is contained in $\mathcal{G}$, not that $\mathcal{G}$ is exactly a half-line.

For four-dimensional Euclidean compact $U(1)$, Guth proves a nonconfining weak-coupling phase for the **Villain** action (reading: abstract); Fröhlich–Spencer establishes a massless Coulomb phase in the setting recorded in B78 (reading: abstract). Those phase results do not determine transition order or a continuously vanishing confined-phase mass. They also do not establish the exact set $\mathcal{G}_{U(1)} = (g_c, \infty)$ for the KS operator defined here; that translation would need its own proof.

For the standard four-dimensional **Wilson** action, numerical evidence supports a **weakly first-order** zero-temperature bulk transition (Vettorazzo–de Forcrand 2004, §§2–3, reading: passage). This replaces the erroneous second-order assertion. The Maxwell description belongs to the weak-coupling massless phase; no Maxwell continuum construction at the transition endpoint is established here. A massless phase over an interval also defeats the original identification of every gapless coupling with a transition point.

The abelian comparison remains useful: group-independent strong-coupling gap arguments alone do not supply weak-side control. The commutator potential of G07, §3, the local-observable restriction of the upper-bound note, Proposition 6, and the formal valley calculation of the valley note, §3b are distinct ingredients with distinct scopes. None is by itself a proof of a nonabelian infinite-volume or continuum gap, or of the global shape of either group’s gapped set.

5. Openness, closedness, and where each stands

For any subset of the connected interval $(0, \infty)$,

$$\mathcal{G} = (0, \infty) \iff \mathcal{G} \neq \emptyset \text{ and } \mathcal{G} \text{ is open and closed.}$$

This is a topological fact, not a regularity theorem for $c(g)$.

property	status	what it would take
$\mathcal{G} \neq \emptyset$	proved by T2: $[g_0, \infty) \subset \mathcal{G}$	strong-coupling theorem
$\mathcal{G}$ open	unproved globally; (g_0, ∞) is a known interior region	volume-uniform stability under changes of g
$\mathcal{G}$ closed	unproved; sufficient for T2' if openness holds	positivity at every finite limit point of gapped couplings, with control preventing lower bounds from collapsing

Correction (2026-10-02): later locality result. The original text left a Lieb–Robinson extension for the unbounded electric terms as the next obstacle. The later locality note, §§1–2 (reading: full-read) treats the commuting on-link electric terms in the interaction picture, leaving bounded finite-range plaquette interactions with unchanged support and norm. This addresses the locality ingredient; no numerical velocity or clustering-length estimate from that note is needed here. Its §5 also explains why locality alone does not prove gap stability: the cited

stability route has additional hypotheses not verified at general finite KS coupling, including frustration-freeness. No universal assertion that every stability theorem requires that hypothesis is intended.

Closedness alone is not T2' without openness, and excluding the two originally proposed scenarios would not establish either property. Finite-volume analytic dependence provides neither estimate. Even a proved T2' would still leave continuum scaling, construction and nontriviality to T3/T4.

6. Consequence for STATE

Correction (2026-10-02). This note supports finite-lattice continuity, the existing nonempty strong-coupling region and a conditional relation between temporal decay and the gap in a specified cyclic vacuum representation. It does not prove an exhaustive infinite-volume gap-closing dichotomy, a phase-transition equivalence, uniqueness of a continuum limit or the exact abelian gapped set. The locality ingredient is supplied by the later note; volume-uniform gap stability remains an estimate to establish. Any use of closing lattice eigenvalues must add limiting-correlator convergence and surviving physical local spectral weight. For the continuum goal, the relevant lower bounds must be carried along the weak-coupling scaling trajectory, with T3/T4; an all-coupling open-and-closed argument is only a stronger sufficient lattice route. No STATE or catalog edit is made in this correction.